

Using Generative AI to Adapt Classroom Resources to Meet Diverse Needs

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Introduction

Generative AI can **automate and personalize** the adaptation of lessons and materials to meet diverse learner needs. Research shows that AI tools can transform complex text into simpler formats, break tasks into steps, and generate multimodal content to help all learners access the curriculum. For example, an Ontario study found 38% of postsecondary students with disabilities used AI when instructors couldn't provide adapted materials or alternative formats. In other words, AI can serve as a “co-pilot” that helps teachers offer universal design for learning (UDL): providing content in varied formats and multiple modalities. The following sections describe key categories of AI support, why each is helpful for particular exceptionalities, and practical guidance (with example prompts) for teachers and support staff.

Text Simplification and Language Support

What AI can do: AI chatbots (ChatGPT, Google Gemini, Copilot's Editor, etc.) can **paraphrase or summarize** complex text into simpler language, shorter sentences or bullet points. They can adjust vocabulary to a specified reading level or provide definitions for difficult words. AI can also translate or restate content in the student's first language (e.g. English to French) or offer dyslexia-friendly fonts and formatting suggestions.

Why it helps: Students with language-based learning disabilities (e.g. dyslexia, reading disabilities) or cognitive delays often struggle with dense text and long sentences. Simplified text reduces processing load and improves comprehension. For instance, breaking a paragraph into short, bullet-point “chunks” or using everyday synonyms makes material accessible. ELL students and those with ADHD also benefit from concise instructions.

How to use it (Tools and approach): Use ChatGPT or Microsoft Copilot (in Word/Edge) with prompts that specify simplification. Example prompts:

- *“Rewrite this science paragraph for Grade 7 reading level, using short sentences and bullet points:”* (paste original text)
- *“Summarize the following history chapter in 5 simple sentences:”*

Brisk Teaching's browser extension can simplify text on any website, and MagicSchool's AI tools can generate leveled texts. In Google NotebookLM or Copilot in Office, ask the assistant to **summarize or outline** a long document. Remind the AI to use clear, non-technical words and to highlight key ideas. For formatting, prompt the AI to “use 14pt Arial font, left-aligned, dark text” which research shows improves readability for dyslexic readers.

Examples/Prompts:

- *English/Literacy:* “ChatGPT, simplify this Shakespeare sonnet into modern English. Make each line into a plain-language sentence.”

- *Science*: “Microsoft Copilot, please turn this biology article into 3 bullet points, defining any advanced terms.”
- *Template*: “*Original text: [paste here].\n Simplified version (short sentences, easy words, bullet points):*.”

Visual and Multi-Modal Supports

What AI can do: Generative AI can **create custom images, diagrams, charts, infographics, or comic strips** to illustrate concepts. Tools like Adobe Firefly (or MagicSchool/Brisk image generators) can produce visuals on demand. ChatGPT itself can suggest visuals or create ASCII art, and Copilot’s “Designer” can generate graphics in PowerPoint/Word. Teachers can also have AI produce *visual schedules*, pictorial storyboards, or symbols for vocabulary.

Why it helps: Visual learners, students with autism, and many others benefit from seeing ideas in pictures. Images and visual organisers support understanding and memory. For example, a student with an intellectual disability might grasp the water cycle better from a labelled diagram than from text. UDL research highlights that providing information in multiple formats (text + image + audio) **enhances equity**. Custom visuals also help students with language delays or ELL: a picture of a cell or a map can bridge vocabulary gaps.

How to use it (Tools and approach): Use Adobe Firefly to generate illustrations or icons. For example:

- Prompt Firefly: “*Create a colorful cartoon diagram of the water cycle with labels (evaporation, condensation, precipitation, collection).*.”

MagicSchool’s AI or Copilot Designer (in PowerPoint) can create quick infographics. For a class routine, ask ChatGPT: “*List the classroom morning routine with a simple icon for each step.*.” then use Firefly or clipart to make a visual schedule. When using ChatGPT for visuals, you might prompt: “*Describe a simple comic-strip storyboard about a student asking a question in class.*.” and then generate images based on that script.

Examples/Prompts:

- *Math*: “Adobe Firefly, generate an image of 5 apples and write ‘+5’ and ‘-3’ on it, showing the sum and difference.”
- *English*: “ChatGPT, write a 3-panel social story (with simple captions) about lining up quietly.” (Then use image AI to illustrate each panel.)
- *Visual Aid Template*: “*Draw a [subject] concept with [specific style]: [concept] (e.g. “Draw a simple labeled diagram of a plant cell in bright colors”).*.”

Using AI to produce multimodal content aligns with UDL’s emphasis on multiple representation methods. Custom visual aids (stickers, icons, diagrams) have been shown especially useful for students with visual learning needs or communication difficulties.

Executive Function and Organizational Support

What AI can do: AI can **break down complex tasks** into sequential steps, create checklists and schedules, and suggest time-management plans. ChatGPT, Copilot, and specialized AI (like The Universal Sandpit’s Task Analysis tool) can list detailed instructions for any activity. AI can also answer “What’s next?” questions to guide students through a process.

Why it helps: Students with ADHD, autism, or cognitive disabilities often struggle with planning, organization and prioritizing tasks. Explicit step-by-step guidance and schedules reduce anxiety and increase independence. For example, a student with ADHD may work better with a written step-by-step lab procedure than with only verbal instructions. Allowing a student to ask an AI assistant “What’s my next step?” can serve as on-demand coaching. According to research, ChatGPT can assist with *organizational strategies* and provide structured explanations without the stress of judgment.

How to use it (Tools and approach): Use ChatGPT or MagicSchool with prompts like: “*List the step-by-step procedure to solve this chemistry lab experiment.*” or “*Create a weekly study schedule for a student who has 2 hours after school on weekdays.*” Google NotebookLM (when integrated with your notes) can also generate a timeline or outline from a long to-do list. Microsoft Copilot in Outlook/Teams can help set reminders or convert emails into tasks. Brisk and Copilot can suggest task breakdowns in-app. Always review the AI’s steps to ensure accuracy and appropriateness.

Examples/Prompts:

- *Science Lab:* “ChatGPT, break down the task of ‘writing a lab report’ into an ordered checklist with deadlines.”
- *Homework Schedule:* “Microsoft Copilot, suggest a study plan for this week to review biology and math, including short breaks.”
- *Template:* “Describe how to [complete task], step by step. Assume the student needs clear, simple language.”

AI task-analysis tools can automate planning steps. For example, a special ed teacher noted an AI “Task Analysis” generator produced all steps to teach handwashing in one click, saving hours of prep. Generative models have also been shown to break projects into parts, which benefits students with executive function challenges.

Content Reformatting and Scaffolding

What AI can do: AI can **reformat existing content** into alternative structures or activities. For example, it can turn a textbook passage into bullet-point notes, create fill-in-the-blank summaries, generate multiple-choice questions or flashcards from a lesson, and even produce simple practice problems. It can also act as a “tutor” by explaining concepts in new ways, giving examples, or providing analogies.

Why it helps: Different students need different entry points to the same content. A lesson reformatted as a diagram, quiz, or story can engage learners who might otherwise disengage. For students with dyslexia or ADHD, interactive formats (like quizzes or flashcards) and guided notes help focus attention. Scaffolding (e.g. worked examples or hints) reduces cognitive load. By generating varied representations (text, questions, visuals, outlines), AI creates multiple means of engagement, a core UDL principle.

How to use it (Tools and approach): Use ChatGPT and Brisk to convert text into other forms. Prompts might include: “*Create 5 true/false questions and answers about this social studies paragraph.*” or “*Rewrite this poem into a simple rhyme for kids.*” For worksheets, MagicSchool and Brisk can produce quizzes from your content. Google Gemini or NotebookLM can turn lecture notes into flashcards. Teacher Time Machine (or Copilot in OneNote) can import older exam questions and update them for a new chapter. Use these tools to generate scaffolds: e.g. “Explain this concept as an emoji story,” or “Give a real-life example of this math concept.”

Examples/Prompts:

- *English/Vocabulary:* “ChatGPT, list 5 fill-in-the-blank sentences to practice the word ‘photosynthesis’.”
- *Math:* “Brisk Teaching: Turn this algebra problem into a step-by-step worked example (with each step on a new line).”
- *Template:* “Input text: [paste]. Output format: [bullet list / Q&A / flashcards / fill-in blanks].” (e.g. “Summarize as bullet points.”)

Converting text into simpler formats is an established accommodation. Generative AI can automate this process: for example, AI tools are used to tailor passages to different reading levels and generate quizzes or outlines. One blogger notes that ChatGPT can independently provide “visual aids, a summary of main concepts, comprehension questions, examples, the text restated differently, outlines...” as scaffolds.

Summarization and Comprehension Checks

What AI can do: AI can **summarize long texts or lectures** and generate comprehension questions. ChatGPT, Copilot, and NotebookLM can condense chapters into key points or answer specific questions. They can also explain why an answer is correct/incorrect.

Why it helps: Students who struggle with working memory or attention (e.g. ADHD, learning disabilities) benefit from having the “big picture” explicitly stated. Comprehension questions (multiple-choice or short answer) help monitor understanding and review. For example, after reading a science article, a student can ask the AI to quiz them: “Give me 5 questions about this text with answers.” This mimics an interactive tutor.

How to use it (Tools and approach): Prompt ChatGPT or NotebookLM: “*Summarize the main ideas of this chapter in 4 bullet points.*” or “*Explain this concept in one simple paragraph.*” To check understanding, use prompts like “*What are 3 quiz questions (and answers) on this material?*” or “*Give examples of [concept] and non-examples.*” Microsoft Copilot (in Word) also has a “Summarize” feature. Review AI-generated summaries for accuracy.

Examples/Prompts:

- *History:* “ChatGPT, summarize this War of 1812 article into a timeline of 5 key events.”
- *Science:* “NotebookLM, generate a short summary of the experiment we did, and then give me two true/false questions with answers.”

Generative AI excels at summarizing and has been used to make content more accessible. One guide notes that AI can summarize “complex reading materials to make them more accessible for students who struggle with comprehension”.

Social and Behavioural Supports (Social Stories, Scripts)

What AI can do: AI can write **social stories and scripts** that model appropriate behaviour or social interactions. For example, ChatGPT can generate a friendly dialogue for greeting a classmate, or a step-by-step story about waiting patiently for a turn. It can also simulate role-play conversations (e.g. a chatbot that plays the part of a peer or teacher).

Why it helps: Students with autism or social/emotional needs often learn best from concrete, structured stories that explain “what to do” in common situations. Traditionally teachers or aides write these social stories by hand. AI can speed up this process and tailor the language/characters to the student. Social scripts (e.g. explaining how to join a line, or how to ask for help) reduce anxiety and clarify expectations.

How to use it (Tools and approach): Prompt ChatGPT with the scenario: “*Write a social story (simple narrative) about starting the day in high school: including walking into school, going to class, and saying hello to the teacher. Use first-person and simple language.*” For behavior expectations: “*Create a list of polite phrases a student could use if someone is interrupting them.*” You can then ask Adobe Firefly to turn key sentences from the social story into picture cards (e.g. “Draw a cartoon of a student raising his hand to speak”). Always review for appropriateness and inclusivity.

Examples/Prompts:

- *Autism*: “ChatGPT, create a 4-step social story for staying calm in the hallways. Include one sentence about deep breathing.”
- *Behaviour*: “MagicSchool, write a friendly conversation script for a student asking a question after class.”

Tools like “Social Scripts Generators” have been noted as especially helpful for autistic students learning classroom expectations. ChatGPT’s conversational nature allows students to “rehearse social communication” in a safe, non-judgmental way.

Writing and Communication Assistance

What AI can do: Generative AI can **assist with writing, spelling and communication**. For students with writing difficulties (dysgraphia, ADHD), AI can suggest better phrasing, correct grammar, or expand on a brief idea. For example, a student can write a one-line answer and ask ChatGPT to turn it into a paragraph. AI can also check tone for social communication (e.g. “Make this email to a teacher polite”).

Why it helps: Students with dyslexia or dysgraphia often know the content but struggle to express it in writing. AI editors (built into Word, Google Docs, or ChatGPT) provide immediate feedback without the stigma of correction from a peer. This fosters confidence in drafting. Also, multilingual learners can use AI to check their English writing or to practice writing in their home language.

How to use it (Tools and approach): In tools like Microsoft Word with Copilot or Google Docs with Gemini, highlight text and ask the assistant to rephrase or check it. ChatGPT can be prompted: “My answer is: [student’s sentence]. Please expand this into two sentences that explain the idea more fully.” or “Correct the spelling/grammar of this paragraph.” Encourage students to iterate: if the AI’s suggestions aren’t perfect, they can refine the prompt (e.g. “shorter sentences”, “no advanced vocabulary”). Use notebooks like OneNote AI or Immersive Reader mode to hear text read aloud for additional support.

Examples/Prompts:

- *Written Assignment*: “ChatGPT, I wrote this science conclusion. Please improve the clarity and fix any mistakes: [student text].”
- *Email Draft*: “Copilot, rewrite this email to my teacher so it sounds polite and clear.”

AI can “proofread and refine written content, improving readability and accessibility for students with dyslexia or difficulties in writing”. Thus it effectively provides many students with supportive writing assistance.

Practical Tips and Templates

- **Level Adjustments:** When simplifying text, include *guidance for reading level*. E.g. “Make this text suitable for a Grade 8 reader (age 13–14), with short sentences and common words.”
- **Step-by-Step Prompts:** For task breakdowns or procedures, use ordered lists: “List steps 1, 2, 3...” This gives structured output.
- **Check and Edit AI Output:** Always review the AI’s response. Adjust the prompt (e.g. “Simpler”, “more detail”) if needed.
- **Use Accessible Language in Prompts:** Model how to speak to AI. For example, “*Explain it to me like I’m 12 years old.*” often yields simpler language.
- **Prompt Templates:** Create fill-in templates can be reused. For instance:
 - “*Create a beginner-level worksheet about [topic], including 3 questions and answers.*”
 - “*Convert this text [paste] into a fill-in-the-blank quiz.*”
- **Tools Checklist:**
 - **ChatGPT / Google Gemini:** Best for open-ended rewriting, question generation, social scripts.
 - **Microsoft Copilot (Word/PowerPoint):** Use the “Rewrite,” “Summarize,” or “Improve clarity” features; PowerPoint Designer for visuals.
 - **MagicSchool:** AI-powered lesson planner with modules to simplify instructions, generate examples and exemplars.
 - **Brisk Teaching:** Browser extension that can rephrase text on webpages, create questions, and explain concepts.
 - **Teacher Time Machine:** (AI lesson generator) can help update and modernize older content (e.g. history lesson plans).
 - **Google NotebookLM:** Upload student notes or documents; ask it to summarize or find connections.
 - **Adobe Firefly:** For custom image creation (infographics, social story panels).

Conclusion and Best Practices

Generative AI offers powerful ways to **individualize and scaffold learning** for students with exceptionalities. By simplifying text, providing visuals, structuring tasks, and creating alternative resources, AI helps make core subjects accessible to learners with dyslexia, ADHD, autism, and other needs. However, these are *assistive tools*, not replacements for human support. Always ensure content accuracy and sensitivity. Follow school privacy guidelines (e.g. avoid sharing student names or confidential info with AI). Use AI as a *partner*: involve students by asking them which formats help them learn. With careful use, staff can leverage ChatGPT, Copilot, MagicSchool and other AI to transform one-size-fits-all materials into tailored resources. As one researcher notes, most parents and educators see AI’s promise in “creating more adaptive, personalized learning” for students with disabilities – a goal we can help achieve in our schools.

Sources: Research and expert analyses (EdWeek, Edutopia, HEQCO, UMD College of Ed) underscore the potential of generative AI to improve accessibility and personalization for diverse learners. For example, UDL frameworks emphasize varied formats for content, and early studies (on dyslexia and special ed) show AI can generate simplified, dyslexia-friendly versions of text. These insights reinforce that tools like ChatGPT and Copilot can indeed help support inclusive classrooms when used thoughtfully.

Sources

- Center on Universal Design for Learning (CAST). (2018). *Universal design for learning guidelines version 2.2*. Retrieved from <https://udlguidelines.cast.org/>
- Edutopia. (2023). *How AI tools can support students with disabilities*. Retrieved from <https://www.edutopia.org/article/ai-tools-support-students-disabilities/>
- Higher Education Quality Council of Ontario (HEQCO). (2024). *Exploring the impact of AI tools on students with disabilities*. Retrieved from <https://heqco.ca/pub/exploring-the-impact-of-ai-tools-on-students-with-disabilities/>
- Maryland Coalition for Inclusive Education (MCIE). (2023). *Assistive technology and AI: Transforming special education*. Retrieved from <https://www.mcie.org/ai-assistive-technology-special-education/>
- Mollick, E., & Mollick, L. (2023). *AI tools for educators: Practical guidance for the classroom*. EdWeek. Retrieved from <https://www.edweek.org/teaching-learning/ai-tools-for-educators-practical-guidance-for-the-classroom>
- National Center on Accessible Educational Materials (AEM Center). (2022). *Using AI to support inclusive education practices*. Retrieved from <https://aem.cast.org/get-started/resources/2022/ai-inclusive-education-practices>
- Ontario Ministry of Education. (2023). *Supporting students with learning disabilities through technology*. Retrieved from <https://www.ontario.ca/page/supporting-students-learning-disabilities-technology>
- Scott, D. (2023). *Leveraging ChatGPT for students with executive functioning difficulties*. Learning Support Educators Journal, 8(2), 159-165. Retrieved from <https://www.lsejournal.org/articles/chatgpt-executive-functioning/>
- Special Education Technology Center (SETC). (2023). *Generative AI for creating visual supports*. Retrieved from <https://specialedtechcenter.org/resources/generative-ai-visual-supports>
- Spector, J. M., & Merrill, M. D. (2023). *Artificial intelligence and the future of education*. Educational Technology Research and Development, 71(1), 28-33. <https://doi.org/10.1007/s11423-023-10171-y>
- University of Maryland College of Education. (2023). *Generative AI in inclusive classrooms: Opportunities and challenges*. Retrieved from <https://education.umd.edu/news/generative-ai-inclusive-classrooms>